teamscale-mcp & teamscale-docs-mcp

gitlab.com/cqse/internal/**teamscale-mcp** · **teamscale-docs-mcp**

Two sidecars: the full REST API plus the Teamscale docs, for any client

Three pieces, three jobs

- 🧩 **Teamscale Claude Code plugin**: curated skills for developers editing code

- 🔌 **teamscale-mcp**: the full REST API as tools (the data)
- 📖 **teamscale-docs-mcp**: the Teamscale docs as tools (the understanding)

> The plugin helps a developer change the code Teamscale analyses. These two help us, and the customers we work with, read, explain, and configure Teamscale itself.
> **This talk is about that second pair.**

Quick recap: published plugin

The official `teamscale/claude-code` plugin:

- Per-user and Claude-Code-only, installed on each workstation
- Bundles curated skills (`fix-findings` , `pr-*` , `local-*`) …
- … plus a local backend (`ts_agent_helper`) that those skills call
- Needs Python, the `teamscale-dev` CLI, a `.teamscale.toml` , and a local checkout

> Its value is **opinionated workflows for a developer fixing findings** in the analysed code, with the repo checked out on their own machine.

What teamscale-mcp is

One central HTTP sidecar, deployed once next to Teamscale:

- *One tool per REST endpoint*, generated at startup from the instance's live OpenAPI spec
- *Per-request identity* via `X-Teamscale-User` / `-Token` headers
- Reachable by *any MCP client* over a URL

> If the API can do it, there's a tool for it, and it always matches the instance's own version.

Configured entirely by env vars

```
teamscale-mcp:

  image: registry.gitlab.com/cqse/internal/teamscale-mcp:latest
  ports: ["8081:8081"]                    # serving: MCP_HOST / _PORT / _PATH, MCP_ALLOWED_HOSTS
  environment:

    # Connection + bootstrap creds (used once, at startup, to fetch the spec)
    TEAMSCALE_SERVER_URL: http://teamscale:8080
    TEAMSCALE_SPEC_USER:  some-technical-user
    TEAMSCALE_SPEC_TOKEN: some-technical-user-api-token
    TEAMSCALE_INCLUDE_INTERNAL: "false"

    # Tool surface: trim to a sharp, read-oriented set (blocklist recommended)
    TEAMSCALE_EXCLUDE_TAGS: "Backup,System,Users,Profilers,SAP,…"
    TEAMSCALE_EXCLUDE_METHODS: "DELETE,PUT,PATCH"
    # TEAMSCALE_INCLUDE_TAGS:  "Findings,Metrics,Test Gap Analysis"
    # TEAMSCALE_INCLUDE_NAMES: "createBaseline"
```

> Per-request identity arrives in the `X-Teamscale-User` / `-Token` headers, never from the environment.

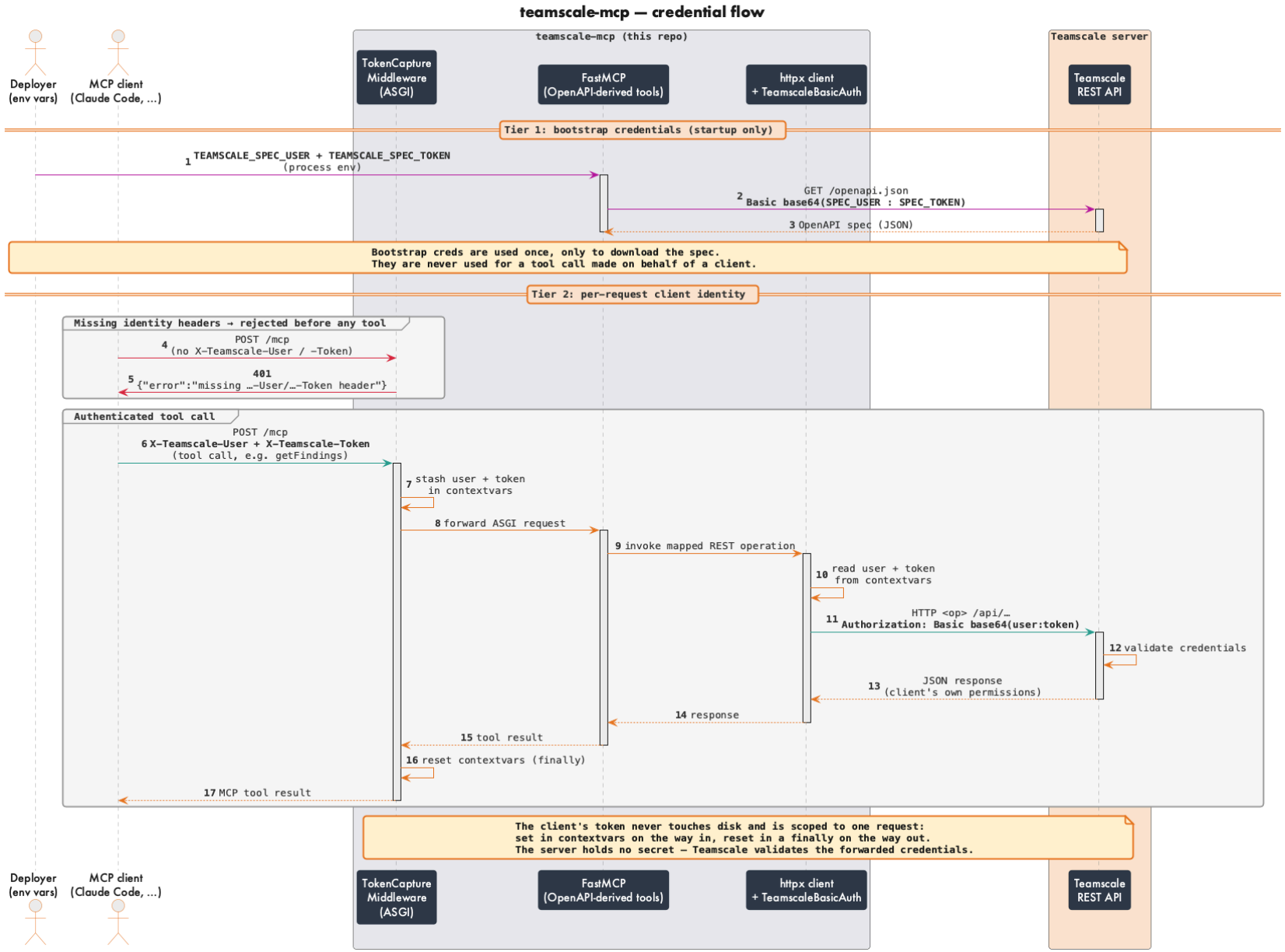
Connect a client with one command

Point any MCP client at the URL, presenting your own identity:

```
claude mcp add teamscale-cqse --scope user --transport http \
  https://cqse.teamscale.io/mcp \
  --header "X-Teamscale-User: bruckner@cqse.eu" \
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

> The two headers *are* your Teamscale identity, forwarded per request.
>
> Any MCP-capable client works the same way.

The credential flow: two tiers, no stored secret



The honest reframe: what it is *not*

It is **not** a drop-in replacement for `ts_agent_helper`:

- Only 3 of 6 skills route through it
- `fix-findings` is a clean win (pure REST)
- `pr-*` is a wash, because the agent has to redo the git/PR resolution the CLI bundled
- `local-*` can't move at all, since they need the local working tree

> "Replaces `ts_agent_helper`" is the weakest way to sell it. The next slides are the strong way.

Where teamscale-mcp really fits

The real value is reach the per-user plugin can't offer:

- 🔎 **Data access** across the entire API, for questions no skill encodes
- 🏛️ **One central deployment** per instance with no per-workstation Python
- 📈 **Observability**: one server, and every tool call is routed through it
- 🤖 **CI and service accounts**: an HTTP endpoint any client can call, as a technical account rather than a person's login
- 🧠 **Any LLM integration**: plain MCP over HTTP, so it's not tied to Claude Code; any MCP-capable client or model can use it
- 👷 **Weakest fit**: an individual dev's daily fix loop, where you'd just use both

A different audience, not just a different deployment

The sharpest distinction isn't *how* it's run, it's *who* it's for.

	Plugin	teamscale-mcp (+ docs)
Works on	the analysed code	the Teamscale data & config
Assumes	a checkout in front of you	just a URL and an identity
Built for	developers editing code	people understanding Teamscale

- Us on customer instances
- The customer managers we report to
- and newcomers setting an instance up

→ People who read Teamscale's results and act on them, without shipping the analysed code.

📖 teamscale-docs-mcp, the understanding half

Teamscale's documentation, served as MCP tools:

- A *sitemap catalog*, plus individual pages converted to Markdown on demand
- *No auth*, because the docs are public content
- Points at an instance's *version-matched bundled docs*, or at the public site
- Self-hosted alongside Teamscale, or run as one central endpoint

> It hands the agent Teamscale's manual, not just its data.

Configured by a single env var

```
teamscale-docs-mcp:

  image: registry.gitlab.com/cqse/internal/teamscale-docs-mcp:latest
  ports: ["8082:8082"]                    # serving: MCP_HOST / _PORT / _PATH, MCP_ALLOWED_HOSTS
  environment:

    # The only real setting: where the docs live. Point at the instance's
    # version-matched bundled docs, so no request leaves the network …
    DOCS_BASE_URL: http://teamscale:8080/documentation

    # … or serve the public site instead:
    # DOCS_BASE_URL: https://docs.teamscale.com
```

No credentials, no headers: the docs are public content, so there is nothing to authenticate.

Explain and act: generic docs, tailored to *your* repo

Neither half is the point on its own. Together they turn a generic manual into steps for *this* repo.

- 📖 **teamscale-docs-mcp** explains: the generic how-to (any language, any stack)
- 🔌 **teamscale-mcp** acts: *this* project's real languages, test framework, build/CI config
- ✍️ Claude merges the two into a runbook written for this repo specifically

> Same how-to page, different repo → a different, correct set of steps every time,
> not a generic doc dump the reader has to adapt themselves.

What "explanation" actually looks like

- 👨‍💼 **A customer's manager**: *"This test gap is 34%. Is that bad, and where did it come from?"*
  The docs explain the concept, and teamscale-mcp pulls the number and trend.
- 🆕 **New to Teamscale**: *"How do I set up an analysis profile for Kotlin?"*
  The docs give the how-to, and teamscale-mcp creates and wires it.
- 👷 **Us, on a customer instance**: *"Compare clone coverage across these five projects."*
  Full REST reach, with the reference at hand, across customer instances.
- 🏗️ **Making sense of a codebase**: *"What's the architecture of my system?"*
  teamscale-mcp queries the code, drafts a Teamscale **architecture**, and sets it up in Teamscale to analyze against going forward.

> Understand first, act second, always grounded in the customer's real data.

A real session: TGA + frontend profiling

Not a mockup, but an actual conversation against a live project:

- Learn what Test Gap Analysis *is* (docs)
- Pull its real number (teamscale)
- Get a setup runbook for frontend coverage (docs), tailored to this project's stack

> Docs and data interleaved end to end.

A real session: TGA + frontend profiling · 1. learn the concept

I want to know about TGA in Teamscale

● I'll look this up in the Teamscale documentation rather than rely on memory. Let me load the documentation tools.

Called teamscale-marcelbruckner-com-docs 4 times

● **TGA = Test Gap Analysis** — one of Teamscale's core "Test Intelligence" features. Here's what it is and how it works.

The problem it solves

Most production bugs occur in **recently changed code**. But testers often don't know everything that changed — refactorings, undocumented bug fixes, or whole features get shipped without the test team being told. The result: risky changes go to production untested. (CQSE's own research found this accounts for the majority of production defects.)

What TGA does

Test Gap Analysis combines three things Teamscale already tracks — **code changes**, **test coverage**, and **issues** — to visualize *every* recently changed method and whether it has been tested yet.

The signature view is the **Test Gap treemap**, where each rectangle is a changed method:
- **Gray** — unchanged code
- **Green** — changed *and* tested
- **Red / orange** — changed but **never tested** → these are your *Test Gaps*

This lets testers spot untested risk and schedule additional test cases *before* release.

Key capabilities

- **Works across all test stages**, not just unit tests: integration, end-to-end, manual, and even hardware/software-in-the-loop tests. Many teams see the true coverage of these non-unit stages for the first time.
- **Issue-level TGA** — aggregates test gaps per Jira issue / Azure DevOps work item, so a tester can see exactly how much of the code changed for a given story/feature is still untested, and filter/prioritize accordingly.
- **Pull-request integration** — Teamscale can post PR comments flagging open test gaps, putting the info directly in front of developers.
- **KPIs & trends** — long-term coverage trend per stage, unaddressed gaps per PR, coverage per user story/defect/epic. Supports a "shift-left" feedback loop.

Getting started (relevant docs)

Task	Page
Record coverage via the profiler	/howto/setting-up-profiler-tga/ (Java, C#, C++, JS, Node, Python…)
Coverage for manual tests	/howto/setting-up-profiler-tga/manual-tests/
Reading the treemaps	/howto/working-with-test-gap-treemaps/
Issue-level test gaps	/howto/analyzing-issue-test-gaps/
Build a TGA dashboard	/howto/creating-a-tga-dashboard/

> *Note: TGA (Test **Gap** Analysis) is distinct from Teamscale's smart **test suggestions** (Test Impact Analysis / Pareto) — those focus on which tests to run fast, whereas TGA focuses on which changes are untested.*

Want me to walk through any specific part — e.g. setting up the profiler to record coverage, reading the treemap, or issue-level test gaps?

✻ Cogitated for 37s

A real session: TGA + frontend profiling · 2. pull the number (first try)

I want to know about TGA in my traureise project

Called teamscale-marcelbruckner-com

● Found three matching projects: traureise, traureise-fastapi-react, and traureise-original. Let me pull the TGA data for the exact traureise project.

Called teamscale-marcelbruckner-com 2 times

● Here's what I found for your **Traureise** project (traureise):

TGA result

Test Gap percentage: 0%

- **Project:** Traureise (traureise), default branch main, configuration completed.
- Computed over the full default-branch history (default baseline, all default coverage partitions).

How to read this

A TGA percentage of **0%** means *zero percent of the analyzed changed code is showing up as a test gap*. There are two very different reasons this can happen, and it's worth confirming which one applies to you:

1. **Everything relevant is covered** — all changed methods in scope have been executed by tests. Genuinely 0 gaps. 🎉
2. **No test coverage has been uploaded** — with no coverage data, there's nothing to compare changes against, so the metric collapses to 0. This is the more common cause for a newer project.

Given the project was created fairly recently and I can't see any coverage partitions referenced, **cause #2 (no coverage uploaded yet) is the likely explanation** — but I'd want to verify rather than assume.

Heads up: three matching projects

There are actually three projects with "traureise" in the name:
- traureise ← this one
- traureise-fastapi-react
- traureise-original

If you meant a different one, tell me and I'll pull its numbers.

Want me to dig deeper?

I can:
- **Check whether coverage exists** (query with all-partitions / execution-only to see if any methods were executed) to confirm which of the two cases above applies,
- **Break down the Test Gap treemap** to see changed vs. tested vs. untested methods,
- **Pull issue-level test gaps** for specific tickets, or
- Run the same for traureise-fastapi-react / traureise-original.

Which would you like?

A real session: TGA + frontend profiling · 3. catching the wrong read

> No, I'm sure coverage exists

● Good — then let me figure out why the default query returned 0%. My first call used only the default coverage partitions; if your coverage lives in other partitions it would be ignored. Let me re-run with all partitions considered, and separately check whether methods are being seen as executed at all.

Called teamscale-marcelbruckner-com 3 times

● All queries — including the churn-only one, which ignores coverage entirely — return 0. That's the key clue: **churn TGA of 0 means Teamscale sees no *changed* methods in the analyzed window.** TGA always measures "of the code that changed since a baseline, how much is untested" — so with no baseline set, the change window is effectively empty and the result collapses to 0 regardless of coverage.

Let me check the configured baselines and re-run over an explicit time window (from before the project's first commit through HEAD).

Called teamscale-marcelbruckner-com 2 times

● That's the answer. The reason every earlier query returned 0% is now clear:

Root cause

Your project has **no configured baselines** (getAllBaselines → empty). TGA measures test gaps *relative to a baseline* (the start of the change window). With no baseline set, the default query had an empty change window → 0 changed methods → 0% by definition. It was never a coverage problem — your coverage is fine, as you said.

Actual TGA for Traureise

Measuring across the **full project history** (from before the first commit through main:HEAD, all coverage partitions):

Test Gap ratio: ~22.1%

Interpreting the treemap semantics:
- **~22%** of the methods changed since the project began have **never been executed by a test** — these are your test gaps.
- **~78%** of changed methods have been covered. 👍

Why this matters for your setup

Because there's no baseline defined, the TGA widgets/dashboards in the UI will also show 0 by default until you give them a time window. Two fixes:

1. **Configure a baseline** (e.g., "last release" or a fixed date) under project settings → then TGA reflects gaps *since that point*, which is the usual way teams track "did we test what we changed this release." See /howto/working-with-test-gap-treemaps/.
2. For ad-hoc analysis, just specify a start/end time window (as I did here) rather than relying on the default.

Want me to:
- Break down the treemap to show **which specific methods/files** make up that 22% gap,
- Compute TGA **since a specific date** (e.g., a recent release) instead of all-history, or
- Check **issue-level test gaps** for particular tickets?

● I've localized the 22% gap down to the file level — that's as granular as the API goes (per-method red/orange cells only exist in the UI treemap). Here's the breakdown.

Where the ~22% test gap lives

Frontend — the bulk of the gap

Path	Test gap
frontend/src (entire tree)	**100%**

None of your frontend source (components/, routes/, hooks/, etc.) maps to any recorded coverage. Even though there are Playwright specs under frontend/tests, no coverage from them is being associated with the frontend source methods in Teamscale — so every changed frontend method counts as a gap. This is the single largest contributor.

Backend — small, concentrated gap (backend/app = 13%)

File	Test gap	Note
backend/app/api/routes/utils.py	**100%**	fully untested
backend/app/utils.py	**14.3%**	~1 in 7 changed methods untested
backend/app/api/routes/login.py	**20%**	e.g. password-recovery / reset handlers typically
backend/app/crud.py	0%	fully covered ✅
backend/app/core/*	0%	✅
backend/app/api/routes/items.py, users.py, private.py, websites.py	0%	✅

Reading this

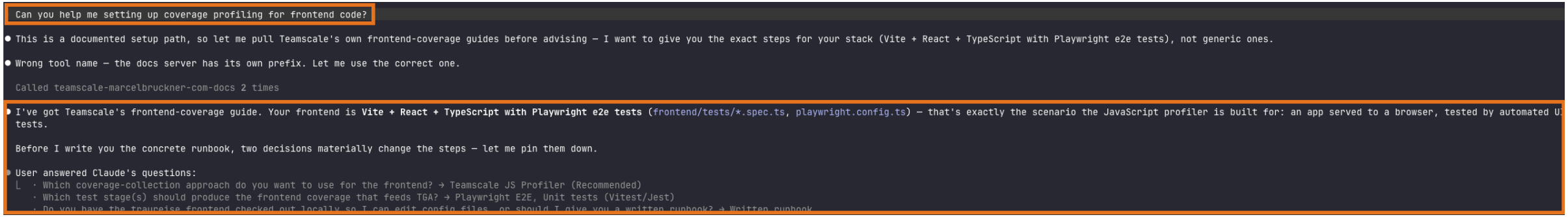
- Your **backend is in good shape** — gaps are limited to utility/helper code (utils.py, the routes/utils.py endpoints) and part of login.py. Those are the concrete methods to add tests for.
- The **headline number is dominated by the frontend**, which shows as 100% gap. Before writing a pile of frontend tests, it's worth checking *why* — most likely the frontend coverage (Playwright/Vitest) either isn't being uploaded or isn't mapped to a coverage partition Teamscale associates with frontend/src. If you expected frontend coverage to exist, that's a coverage-upload/mapping issue rather than a genuine testing gap.

Getting exact method names

The public API exposes TGA only as a percentage per path, not the individual red/orange methods. To see the exact untested **methods**, open the **Test Gap treemap** in the Teamscale UI for traureise (set the same time range / a baseline) — each red or orange cell is a specific changed-but-untested method, and clicking it jumps to the code.

Want me to:
- Drill the frontend into subfolders (components vs routes vs hooks) to rank them, or
- Check whether **any** frontend coverage partition exists at all (to confirm the upload/mapping theory)?

A real session: TGA + frontend profiling · 5. ask for a fix



● Here's your runbook. Since you want **both** stages, note one important split up front:

- **Playwright E2E** runs in a real browser → use the **Teamscale JavaScript Profiler** (instrumenter + collector). This is the recommended path.
- **Vitest unit tests** run in Node/jsdom → the profiler *can't* be used there (instrumented code fails in Node). Vitest produces its own coverage, which you upload separately.

So you'll end up with **two coverage partitions** feeding TGA, e.g. E2E Coverage and Unit Test Coverage.

Part A — One-time setup in Teamscale (both stages)

1. Go to **Project Configuration → Coverage Profilers** in the traureise project.
2. Create a profiler config, ID traureise-e2e:

```
teamscaleProject=traureise
teamscalePartition=E2E Coverage
dumpAfterMins=120
```

2. (The Vitest path doesn't need a profiler config — it uploads directly via teamscale-upload.)

Part B — Playwright E2E coverage (JS Profiler)

1. Enable source maps (so dist/ maps back to frontend/src)

frontend/tsconfig.json:

```
{ "compilerOptions": { "sourceMap": true, "inlineSources": true } }
```

frontend/vite.config.ts:

```
export default defineConfig({
  build: { sourcemap: true },
})
```

2. Build and instrument the app

The profiler instruments the **built** output — you must serve the built+instrumented app, *not* the Vite dev server.

```
cd frontend
npm run build                       # produces ./dist with source maps

npx @teamscale/javascript-instrumenter \
  --input ./dist/ \
  --collector localhost:54678 \
  --config-id traureise-e2e \
  --commit "$(git rev-parse HEAD)" \
  --in-place \
  --include-origin 'src/**/*'
```

- --commit ties the coverage to the exact revision being tested (essential for TGA).
- --include-origin 'src/**/*' matches your frontend/src sources.

3. Start the collector (auto-upload to Teamscale)

Add to frontend/package.json:

```
"scripts": {
  "collector": "coverage-collector --teamscale-server-url https://teamscale.marcelbruckner.com --teamscale-user <user> --teamscale-access-key <key> --teamscale-project traureise",
  "pree2e:cov": "npx pm2 delete CC; npx pm2 start npm --name CC -- run collector",
  "e2e:cov": "playwright test",
  "poste2e:cov": "npx pm2 delete CC"
}
```

Install: npm install -D @teamscale/coverage-collector pm2

4. Point Playwright at the instrumented build

In playwright.config.ts, make webServer serve the built dist/ (e.g. vite preview or npm run preview), not vite dev:
webServer: { command: 'npm run preview', url: 'http://localhost:4173', reuseExistingServer: false }

5. Content Security Policy (only if you set one)

The FastAPI full-stack template doesn't ship a strict CSP by default, so the WebSocket to the collector should just work. If you later add a CSP, it must allow the collector:
connect-src 'self' ws://localhost:54678;
script-src 'self' blob: ws://localhost:54678;
worker-src 'self' blob: ws://localhost:54678;

6. Run it

npm run e2e:cov
The collector receives coverage over WebSockets as Playwright drives the browser, then uploads to the E2E Coverage partition.

Part C — Vitest unit-test coverage

> *If you don't have a Vitest suite yet, this stage produces nothing until you add one. Say the word and I'll help scaffold it.*

1. Enable coverage in Vitest

```
cd frontend
npm install -D @vitest/coverage-istanbul
vite.config.ts (or vitest.config.ts):
test: {
  coverage: {
    provider: 'istanbul',
    reporter: ['lcov'],          // lcov → Teamscale reads it directly
    reportsDirectory: './coverage',
    include: ['src/**/*'],
  },
}
```

2. Run and upload

```
npx vitest run --coverage
npx @cqse/teamscale-upload \
  --server https://teamscale.marcelbruckner.com \
  --project traureise \
  --user <user> --accesskey <key> \
  --partition "Unit Test Coverage" \
  --format LCOV \
  --commit "$(git rev-parse HEAD)" \
  ./coverage/lcov.info
```

Part D — Verify

After each upload, TGA for the frontend should stop reading 100%. You can confirm by asking me to re-run getTgaPercentage for frontend/src (all partitions), or check the Test Gap treemap in the UI.

A couple of things worth flagging:
- **The instrumenter must run against the built bundle each time you test a new commit** — wire steps B2–B6 into your CI so coverage stays current, otherwise TGA drifts stale.
- **Use git rev-parse HEAD consistently** for --commit in both stages so E2E and unit coverage land on the same revision.

Want me to (a) generate a ready-to-paste CI job (GitHub Actions, matching your repo layout), or (b) help scaffold the Vitest suite if you don't have one yet?

The same pairing, on a tool outside either server

`teamscale-docs-mcp` also documents `teamscale-dev`.

> `teamscale-dev` itself stays out of scope for both servers, but *explaining* it doesn't.

The same pairing, on a tool outside either server

```
Check the teamscale docs. How do i configure teamscale-dev

● I'll search the Teamscale docs for how to configure teamscale-dev.

  Called teamscale-marcelbruckner-com-docs 2 times

● Found it: /reference/cli/teamscale-dev/. Let me read that page.

  Called teamscale-marcelbruckner-com-docs

● Let me also pull the .teamscale.toml reference, since that's the config file teamscale-dev reads.

  Called teamscale-marcelbruckner-com-docs

● Here's how to configure teamscale-dev (the Teamscale developer CLI). There are two distinct pieces: what to analyze (a .teamscale.toml file) and how to authenticate (server credentials).

  1. Project config: .teamscale.toml

  teamscale-dev reads .teamscale.toml files (TOML format) to know which server/project/branch a directory maps to. A minimal file:

  server.url = "https://teamscale.io/"
  project.id = "example"

  Full set of properties:

  version = "1.0"   # optional, default 1.0
  root = false      # if true, config discovery stops here and doesn't look further up

  [server]
  url = "https://teamscale.io/"   # must resolve in the merged config; must match a server you have credentials for

  [project]
  id = "example"        # the Project ID (may differ from the display name)
  branch = "main"       # optional; if unset, discovered from VCS, falling back to server default
  path = "remote/path"  # optional; path within the Teamscale project ("" = project root)

  Notes:
  - Merging: files inherit from parent directories (like .editorconfig). Put server.url + project.id in your repo root, and override just project.path in subdirectories. A root = true file stops upward discovery.
  - Auto-generate it: in Teamscale, go to Project Configuration → Projects and click Configure IDE plug-ins / CLI to download a ready-made file.
```

The same pairing, on a tool outside either server

2. Credentials (username + access key)

Config files hold no secrets — you supply your username and access key separately. Five ways, in precedence order:

Method	How
`--server` option	`--server https://user:accesskey@example.com/teamscale/` (can repeat per server)
`TEAMSCALE_DEV_SERVERS` env var	Same credentials-URL format, space-separated for multiple servers
`--user` / `--accesskey` options	`--user NAME --accesskey KEY` (applies to all servers not matched by `--server`)
`TEAMSCALE_DEV_USER` / `TEAMSCALE_DEV_ACCESSKEY` env vars	Env equivalent of the above
Args file	Put `--server ...` in `~/.teamscale-dev.args`, then pass `@"${HOME}/.teamscale-dev.args"`

Recommended for terminal use — set once in your `~/.zshrc` (you're on zsh/macOS):

```
export TEAMSCALE_DEV_SERVERS="https://username:accesskey@example.com/teamscale/"
```

Then you never need credential flags. ⚠️ Avoid passing `--accesskey`/`--server` directly on the command line in shared environments — other users can read process args (`ps`). Prefer env vars or the args file. Note `@` in a username must be percent-escaped as `%40` (though `teamscale-dev` auto-corrects `@`).

Verify it works

```
teamscale-dev verify-configuration src/   # validates .teamscale.toml + server/project access
```

Common commands once configured

- `teamscale-dev pre-commit src/` — upload local changes for pre-commit analysis and show findings (add `--upload-uncommitted-files` when paths are under Git/SVN — recommended)
- `teamscale-dev fetch-findings src/` — show findings already on the server (no upload)
- `teamscale-dev open-in-browser src/` — open the file/dir in the Teamscale web UI
- `teamscale-dev mcp` — run as an MCP server for AI coding assistants
- `teamscale-dev generate-completion` — shell tab-completion (bash/zsh)

Other useful flags: `--branch` (override branch), `--insecure` or a `#trust-all-certificates` URL suffix (disable TLS validation — testing only), `--debug`/`--trace` (logging).

Want me to help you write a `.teamscale.toml` for a specific project or set up the zsh env vars?

The same pairing, on a tool outside either server

and why do i need user and accesskey then

● Good question — you **don't** need both. They're alternative ways to supply the same credentials, useful in different situations.

The distinction is **how credentials get matched to a server URL**:

- `TEAMSCALE_DEV_SERVERS` (and the `--server` flag) — credentials are *bound to a specific server URL*, because the username and access key are baked into the URL itself (`https://user:key@example.com/...`). This is per-server.
- `TEAMSCALE_DEV_USER` / `TEAMSCALE_DEV_ACCESSKEY` (and `--user` / `--accesskey`) — credentials that are *not tied to any particular URL*. They apply as a fallback to **every** server referenced in your `.teamscale.toml` files that doesn't already have credentials from a `--server`/`TEAMSCALE_DEV_SERVERS` entry.

So:

Situation	What to use
One server, or same login everywhere	Either one alone is enough — `TEAMSCALE_DEV_SERVERS` is fine
Different logins for different servers	`TEAMSCALE_DEV_SERVERS` (each URL carries its own credentials)
One default login for whatever server the config points at	`TEAMSCALE_DEV_USER` + `TEAMSCALE_DEV_ACCESSKEY`

If you've set `TEAMSCALE_DEV_SERVERS` for your server, you do not also need `TEAMSCALE_DEV_USER`/`TEAMSCALE_DEV_ACCESSKEY`. The `--server` form takes precedence anyway.

Why offer the `--user`/`--accesskey` form at all? Because it separates the *URL* (which lives in the shared, committed `.teamscale.toml`) from the *credentials* (which are personal). With it you can keep just `server.url = "https://example.com/teamscale/"` in the project config and let each developer supply their own user/key via env vars — no secrets in the URL, no secrets in version control.

So for your single-server case: pick one approach. `TEAMSCALE_DEV_SERVERS` alone is all you need.

The honest ledger: capability, not opinion

✅ Can

- Expose the **entire REST API** as tools
- Serve **any MCP client** (Claude Code, CI, web, cron)
- Authenticate as **any identity** (personal or service)
- **Explain and act** together (with docs-mcp)
- **Observability**: every call through one vantage point
- **Retrofit** old instances

🚫 Can't

- **No local working tree** → `local-*` stay on `teamscale-dev`
- **Raw tools, not workflows** → orchestration on the agent, so not the smoothest fix/PR loop

> It's an addition, not a replacement for `ts_agent_helper`

Future work: from prototype to product

🧪 Today — a prototype

- A **working proof of value**, not a finished product
- Shows how much an agent gains with Teamscale in reach: **query and act** (`teamscale-mcp`) and **understand in context** (`teamscale-docs-mcp`)
- Runs as a **sidecar**, bolted on from the outside

🚀 Next — part of Teamscale

- The value is proven; the **home is inside the product**
- A **first-class, native** MCP capability
- **Shipped, versioned, and supported** with Teamscale

> We see a lot of **value for our customers** here, and we'd love to see this become a part of Teamscale.

Thank you

teamscale-mcp gives Teamscale's data to any agent.
teamscale-docs-mcp gives it the manual.

Questions?

Backup

Connect multiple clients with one command each

Point any MCP client at the URL, presenting your own identity:

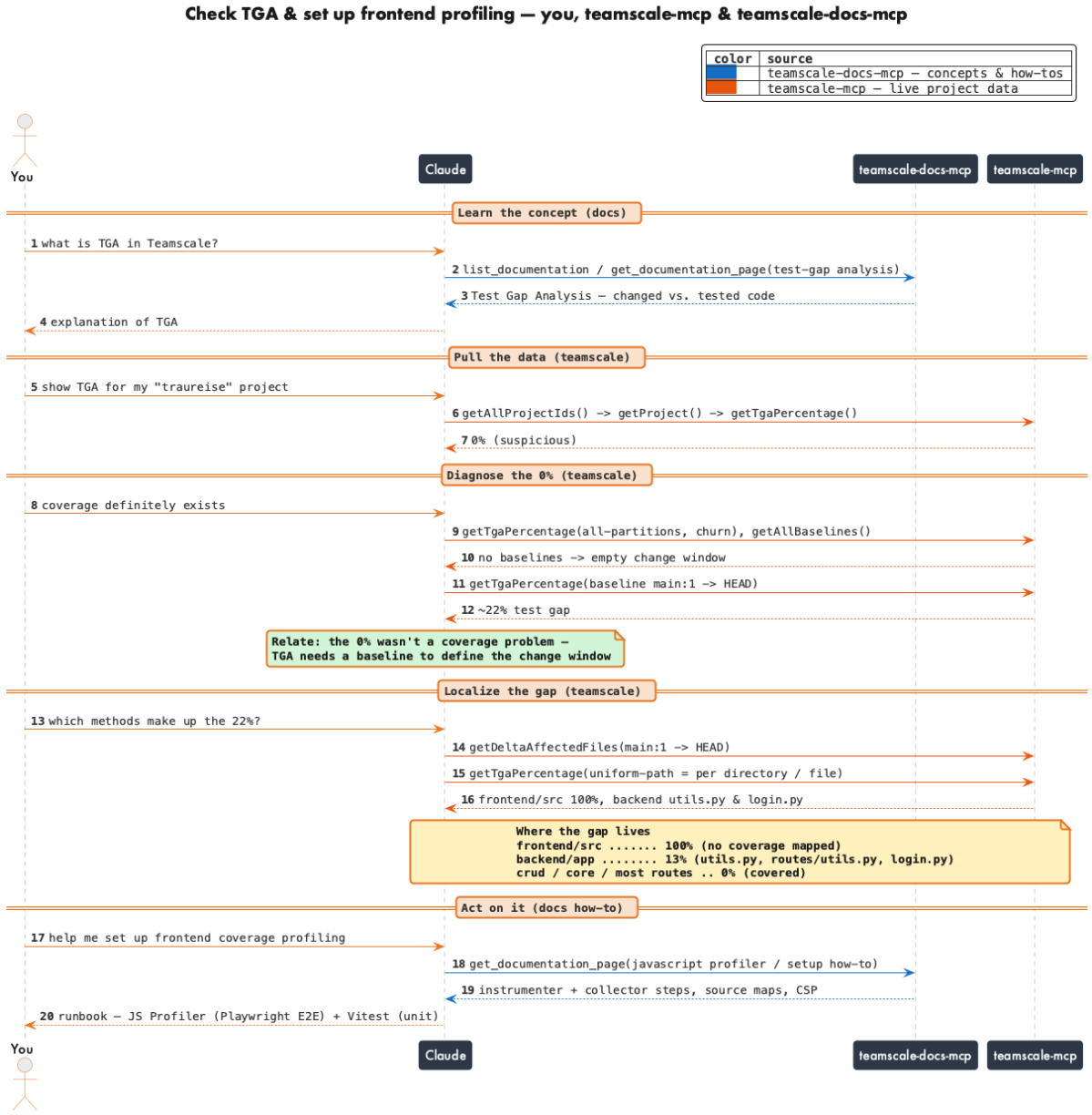
```
claude mcp add teamscale-cqse --scope user --transport http \
  https://cqse.teamscale.io/mcp \
  --header "X-Teamscale-User: bruckner@cqse.eu" \
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

And add as many servers as you need:

```
claude mcp add teamscale-personal --scope user --transport http \
  https://teamscale.marcelbruckner.com/mcp \
  --header "X-Teamscale-User: marcel" \
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

> The two headers *are* your Teamscale identity, forwarded per request.
>
> Any MCP-capable client works the same way.

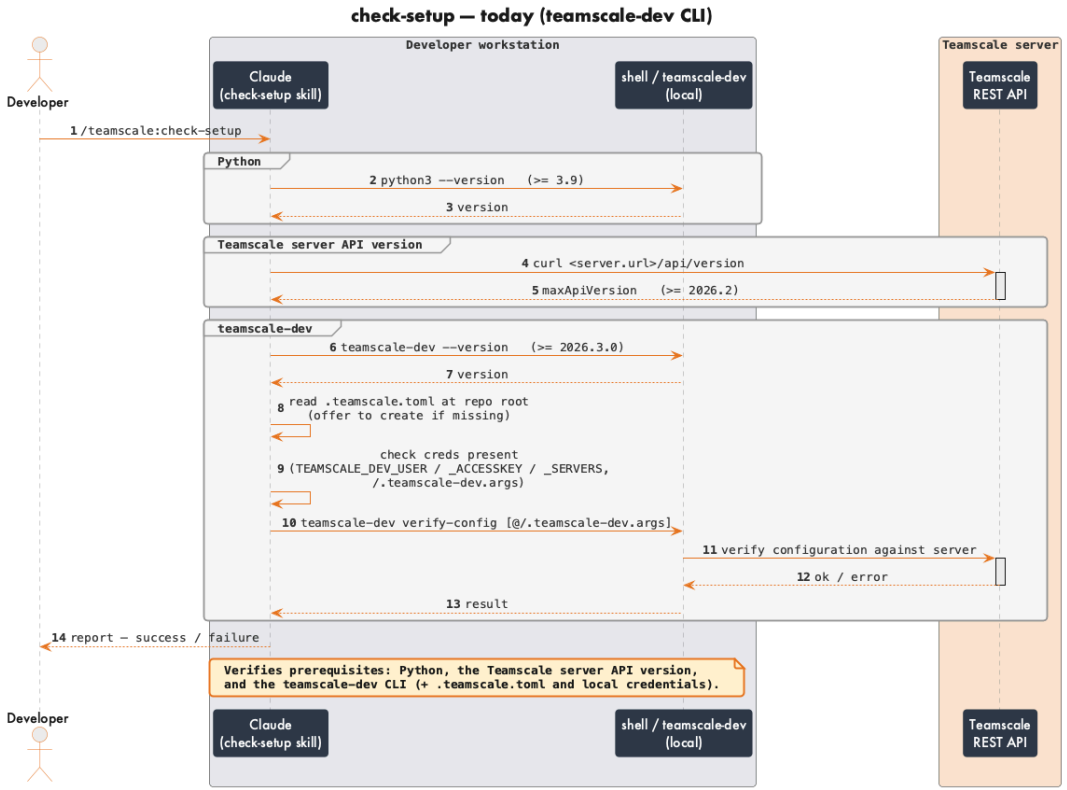
A real session: TGA + frontend profiling

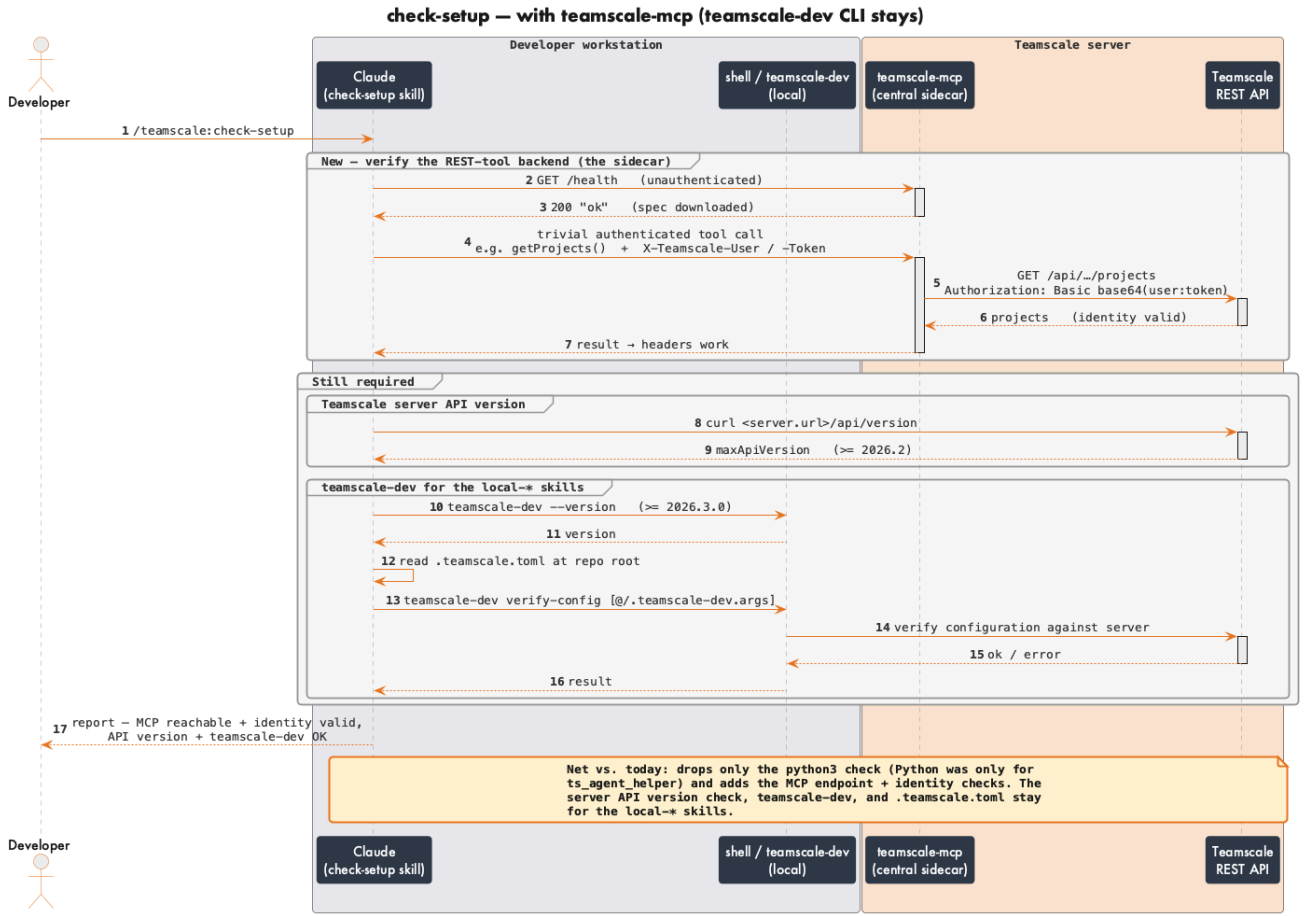


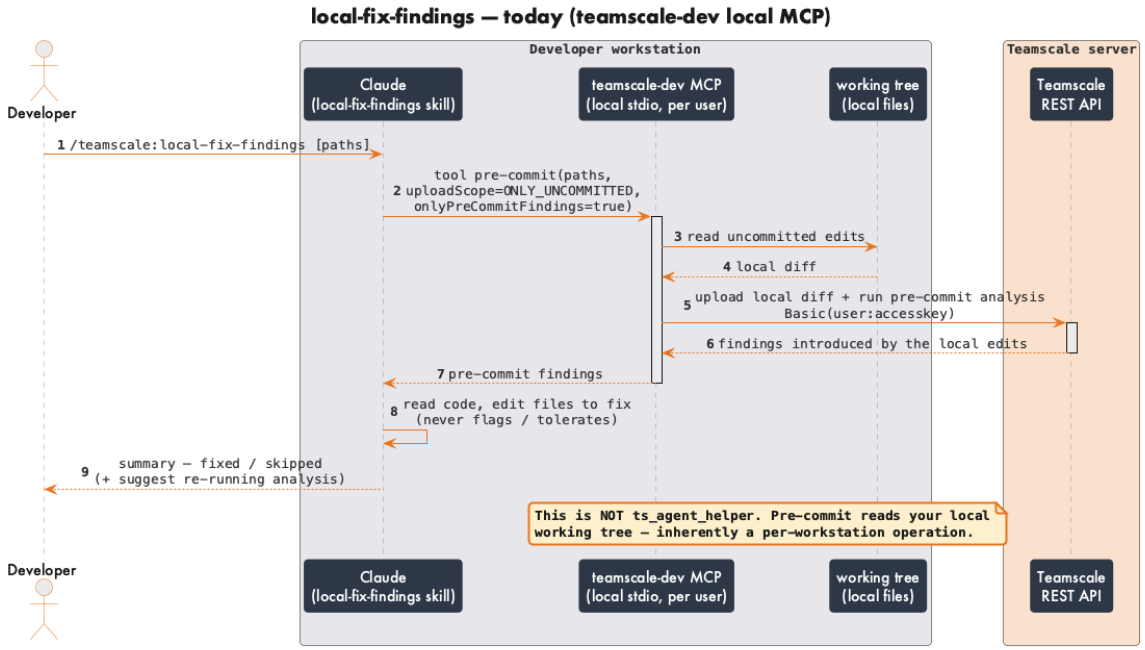
Plugin comparison

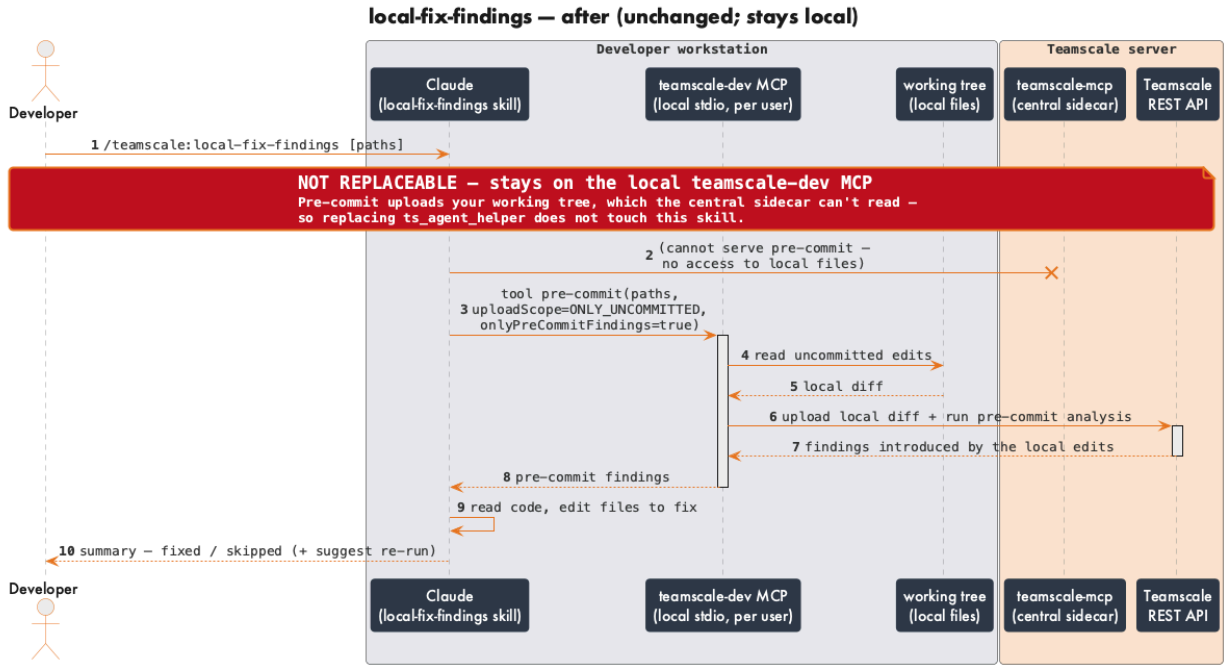
Each diagram: **today (** `ts_agent_helper` **)** on the left, **with** `teamscale-mcp` on the right.

check-setup — shifts to verifying the endpoint

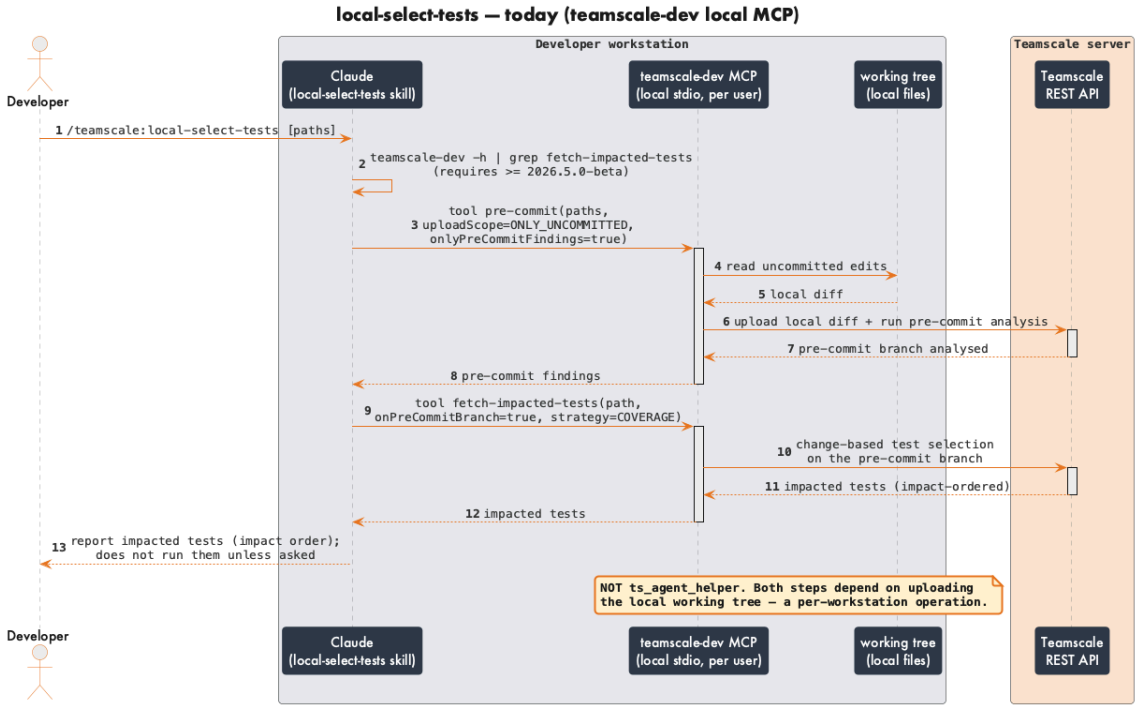


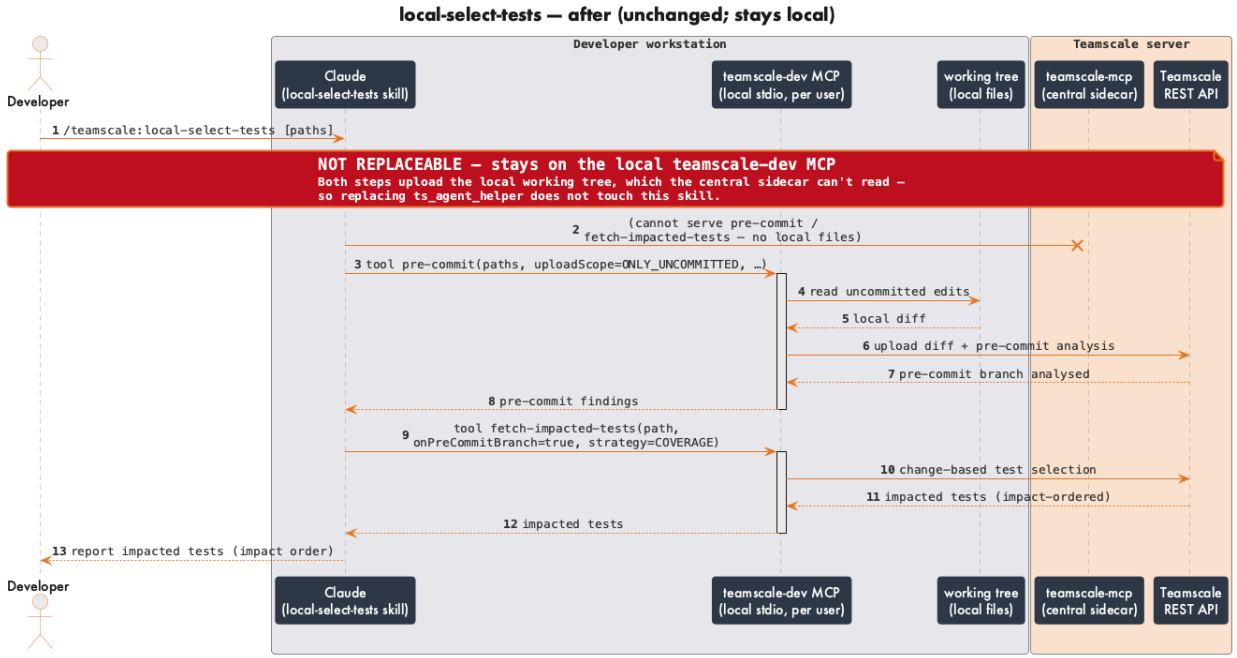



local-fix-findings — today (teamscale-dev local MCP)
Developer workstation
Teamscale server
Developer
Claude (local-fix-findings skill)
teamscale-dev MCP (local stdio, per user)
working tree (local files)
Teamscale REST API
1 /teamscale:local-fix-findings [paths]
2 tool pre-commit(paths, uploadScope=ONLY_UNCOMMITTED, onlyPreCommitFindings=true)
3 read uncommitted edits
4 local diff
5 upload local diff + run pre-commit analysis Basic(user:accesskey)
6 findings introduced by the local edits
7 pre-commit findings
8 read code, edit files to fix (never flags / tolerates)
9 summary — fixed / skipped (+ suggest re-running analysis)
This is NOT ts_agent_helper. Pre-commit reads your local working tree — inherently a per-workstation operation.

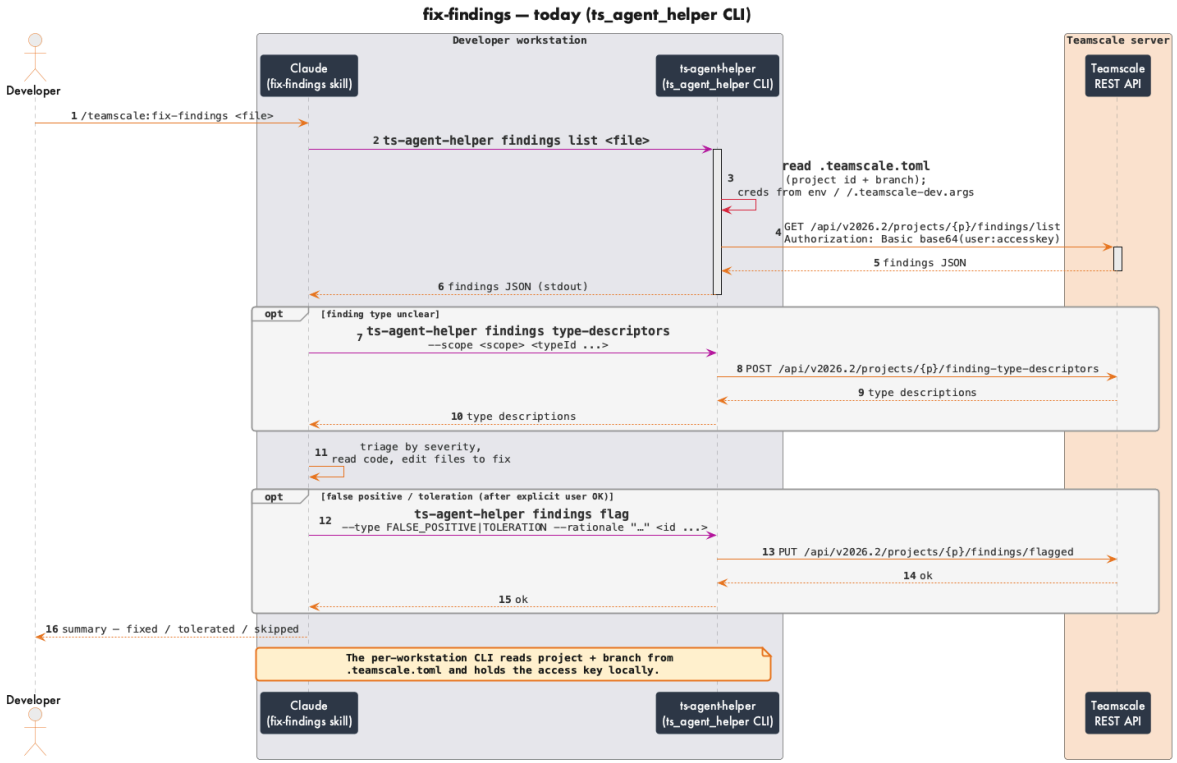

local-fix-findings — after (unchanged; stays local)
Developer workstation
Teamscale server
Developer
Claude (local-fix-findings skill)
teamscale-dev MCP (local stdio, per user)
working tree (local files)
teamscale-mcp (central sidecar)
Teamscale REST API
1 /teamscale:local-fix-findings [paths]
NOT REPLACEABLE — stays on the local teamscale-dev MCP
Pre-commit uploads your working tree, which the central sidecar can't read — so replacing ts_agent_helper does not touch this skill.
2 (cannot serve pre-commit — no access to local files)
3 tool pre-commit(paths, uploadScope=ONLY_UNCOMMITTED, onlyPreCommitFindings=true)
4 read uncommitted edits
5 local diff
6 upload local diff + run pre-commit analysis
7 findings introduced by the local edits
8 pre-commit findings
9 read code, edit files to fix
10 summary — fixed / skipped (+ suggest re-run)

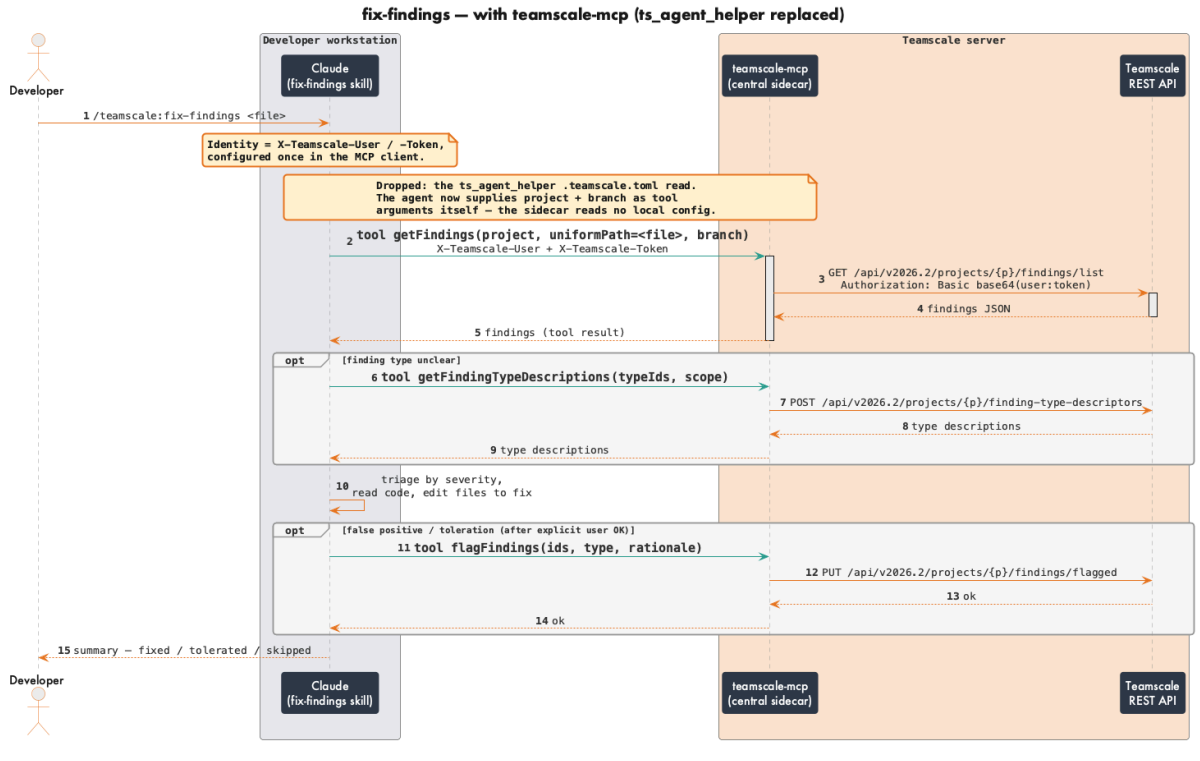
local-select-tests — stays local



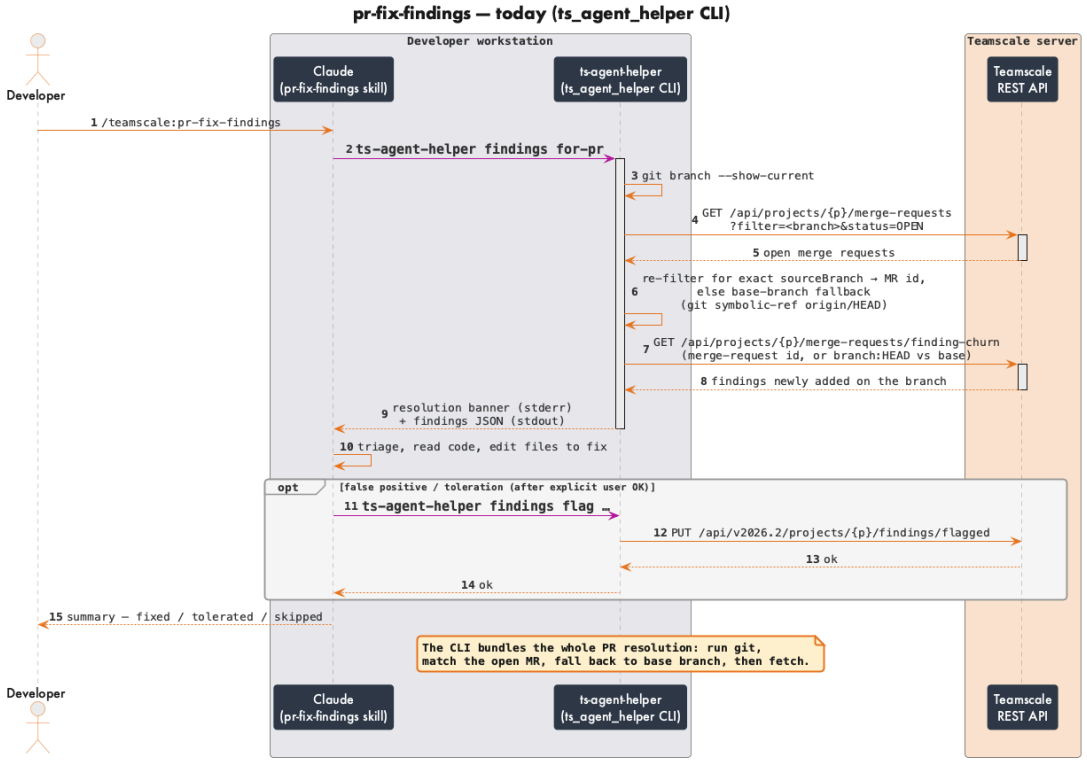


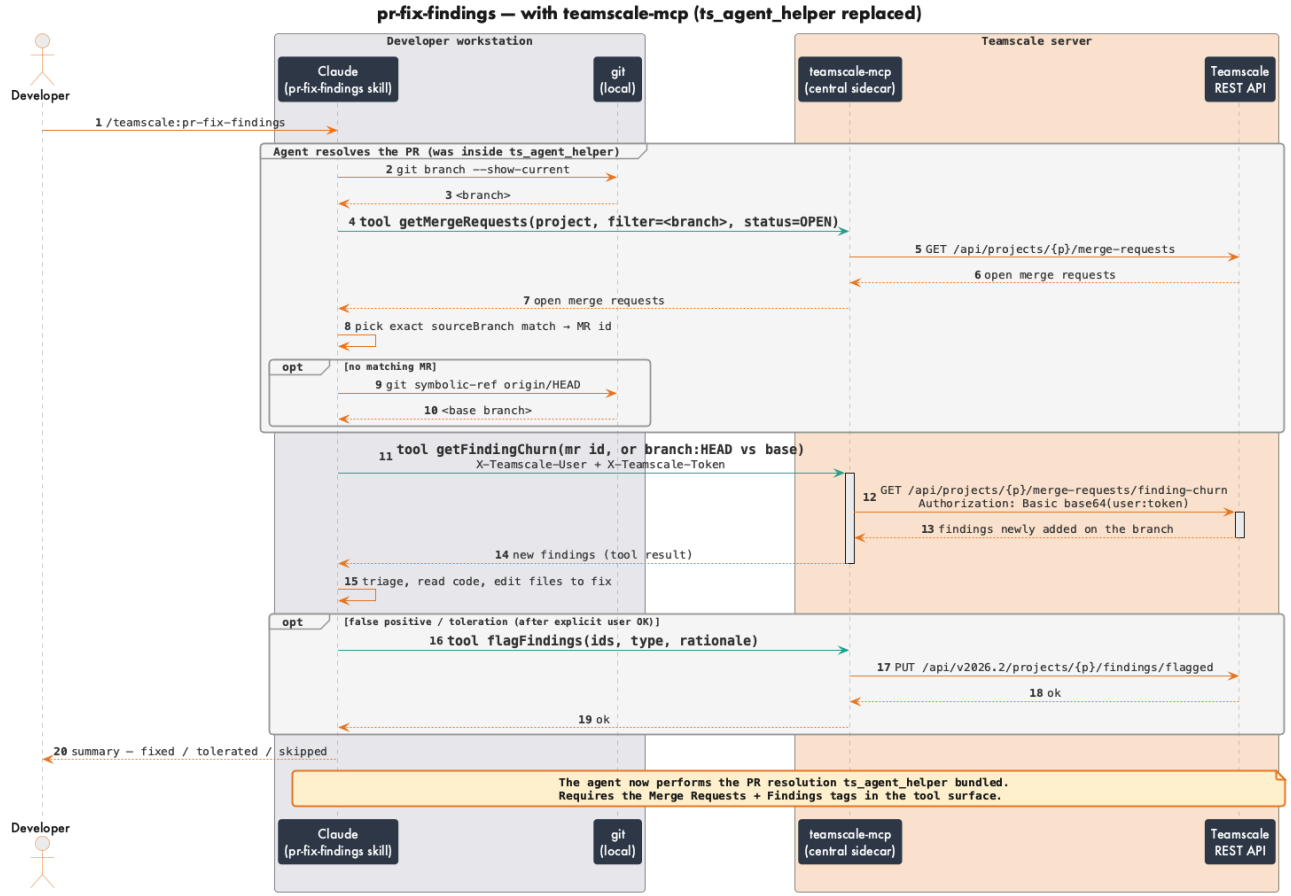
fix-findings — replaceable (clean win)





pr-fix-findings — replaceable (agent takes over PR resolution)





pr-close-test-gaps — replaceable (agent takes over PR resolution)